

JUNIOR SCIENTIST SHOWCASE

Discovering Science Right at Home!

Hey There, Little Scientist! Welcome to your science fair journal. Science isn't just trapped inside big school textbooks—it is a magical superpower hidden inside your kitchen, your backyard, and everything you see, touch, and do! In this journal, you will learn to explore mini-mysteries, test out wild ideas, and think like a real science detective.

The Kitchen Counter Mystery:

Have you ever seen something at home fizz up, float out of nowhere, or instantly scatter away? That isn't magic—it is real chemistry and physics in action! A science fair is a special space where you get to investigate these mysteries firsthand. Instead of just watching, you discover the exact secret reasons **why** things happen!

MEET THE 5 SCIENCE DETECTIVE STEPS

Scientists all over the world use a grand puzzle-solving method to test their ideas. We call them the **5 Detective Steps**. They guide you step-by-step so you never get lost!

1. **Look Closely (Observe):** Use your eyes and senses to notice cool changes in the world.
2. **Ask a Question:** Choose a specific secret mystery you want to unlock.
3. **Make a Smart Guess (Hypothesis):** Predict what you think will happen before you test it.
4. **Try It Out (Experiment):** Set up a fun, safe, and completely fair test.
5. **Tell the World (Share):** Present your awesome results on a colorful poster board!

 **Our Science Slogan: "Look, Ask, Guess, Try, Tell!"**

NEW HOME EXPERIMENTS TO TRACK THE STEPS

Step 1: Look Closely

Example 1: The Soap & Pepper Scatter Trick

How to set it up: Fill a shallow bowl with clean water. Sprinkle a layer of powdery black pepper across the top surface. The pepper flakes will sit quietly, floating like tiny boats on the water.

The Experiment: Dip the very tip of your clean finger into a tiny droplet of liquid dish soap. Now, gently touch the exact center of the water bowl with that soapy finger.

What to Observe: *Swoosh!* Instantly, all the tiny pepper flakes will sprint away to the outer edges of the bowl as if they are scared of your finger! This happens because the soap breaks the water's invisible "skin" (surface tension), pulling the water and the pepper outward rapidly.

Step 2 & 3: Ask & Make a Smart Guess

Example 2: The Floating Egg Race

Once you see items float or sink, your brain naturally creates a new question. Let's make a smart guess using an egg!

The Setup: Imagine you have two identical cups of water.

- **Cup A:** Plain, clear tap water.
- **Cup B:** Water mixed with 3 big spoonfuls of salty white salt.

Forming Your Smart Guess: What will happen if you gently lower a fresh egg into each cup? Salt makes water much heavier and denser, which helps push things upward. Your formal smart guess sentence should sound like this:

"If I put an egg in salty water, THEN it will float to the top because salt water is stronger!"

Step 4: Try It Out & The "Fair Test" Secret Rule

Example 3: The Color-Changing Celery Journey

When you perform an experiment, you must follow the absolute golden rule of science: **You are only allowed to change ONE secret thing at a time!** This ensures your test is completely fair.

Meet Pinky's Unfair "Cheating" Plant Test:

Our friend Pinky wanted to find out if green celery stalks can drink colored water. She took two stalks:

- **Celery Stalk A:** Put in a warm, sunny room inside a cup of clear water.
- **Celery Stalk B:** Put in a dark, cold refrigerator inside a cup of bright red water!

What went wrong? Pinky changed TWO things: the temperature/sunlight AND the color of the water! If Celery B changes color or looks wilted, we won't know if it's because of the cold fridge or the red dye.

Fixing it to make a Fair Test: Keep both Celery stalks side-by-side on the exact same sunny kitchen counter! Only change the liquid: clear water in one, and red water in the other. Within a few hours, you will see the red color climb up into the tiny leaves of Stalk B like magic, showing exactly how plants drink!

STEP 5: TELL THE WORLD! (YOUR POSTER)

After you complete your fair test, you possess exclusive scientific knowledge! It is time to organize your hard work on a big cardboard display board to share with your family and teacher:

- **The Top Sky:** Write your project's title in huge, colorful, bubbly letters (e.g., *"THE SCATTERING PEPPER MYSTERY!"*).
- **The Middle Canvas:** Stick real photos or draw simple crayon pictures of your bowls, cups, eggs, or colored celery stalks.

- **The Bottom Corner:** Write down your final results and proudly declare whether your smart guess was right or wrong!



OFFICIAL JUNIOR SCIENTIST MISSION CHECKLIST

Are you ready to claim your title? Complete this at-home checklist to earn your scientist badge!

- ☐ **LOOK:** Find 3 items in your kitchen (like milk, honey, salt, or oil).
- ☐ **ASK:** Pick a safe question with an adult (e.g., "Does honey mix faster in hot or cold water?").
- ☐ **GUESS:** Write your "If... then..." sentence in your journal.
- ☐ **TRY:** Run a Fair Test, making sure to change only ONE item at a time.
- ☐ **TELL:** Sketch your final experiment on paper and show it to your teacher!

CONGRATULATIONS! Keep exploring and asking big questions!

MY AT-HOME SCIENCE FAIR JOURNAL LOG

My Step	My Creative Lab Notes (Draw or Write Here!)
1. What I Looked At:	
2. My Question:	
3. My Smart Guess:	"IF _____, THEN _____!"
4. My Fair Test:	What I kept the same: The ONE thing I changed:
5. What Happened!	